Each block below names one statement from the paper, shows the computation that tests it, and lists the numbers obtained. Statements that are equalities or inequalities were tested directly, usually to machine precision or in 60-digit arithmetic where double precision could not resolve the sign. Steps that rest on structural theorems (Nash and analytic curve selection, Noetherianity, the Cheng–Di Scala–Yuan finite-sum theorem) cannot be verified by computation; for those the mechanism is illustrated and the block is marked accordingly. No computation contradicted any step.

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20 statements verified numerically **1** structural step illustrated only **21** figures

0 contradictions found

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§1–2 Variational structure

- Energy bounds (2.2) and the even-exponent minimizer
- Proposition 2.2: $0 < p < 2$ forces an orthonormal basis
- Even exponents: averaging over O(d) is a projection
- Lemma 2.1: contact and centered positivity
- Proposition 2.3: the global variational bound (2.3)

§3 Rational exponents

- Lemma 3.1 and the rank bound (1.2)/(3.1)
- Theorem 1.1(1): finite minimizing replacement
- Lemma 3.2: finite-dimensional vanishing orders
- (3.3)–(3.4) and sharpness of the denominator obstruction

§4 Analytic rigidity and great circles

- Theorem 1.2: finite rank ⇔ finite support at irrational p
- Theorem 4.1 (noninteger p): divided differences and the sign obstruction
- Theorem 4.1 (odd p): the trigonometric zero count

§5 Contact curvature

The Hessian formula (5.1) and strict positivity at contacts

The exact centered split (5.2)–(5.4)

Lemma 5.1 and Proposition 5.3: continuity, Hölder constant, separation, packing

§6 Finite-support regularization

Theorem 1.4 and Lemma 6.1: the flat-diagonal correction

Proposition 6.2: maximal-purity selection

Proposition 6.3: the rotationally invariant counterexample

§7 Approximation, selection, recovery

The exponent-Lipschitz estimate (7.1)

Proposition 7.1: one-sided selection and the derivative formula

Theorem 7.2 and Corollary 7.3: target-centered recovery

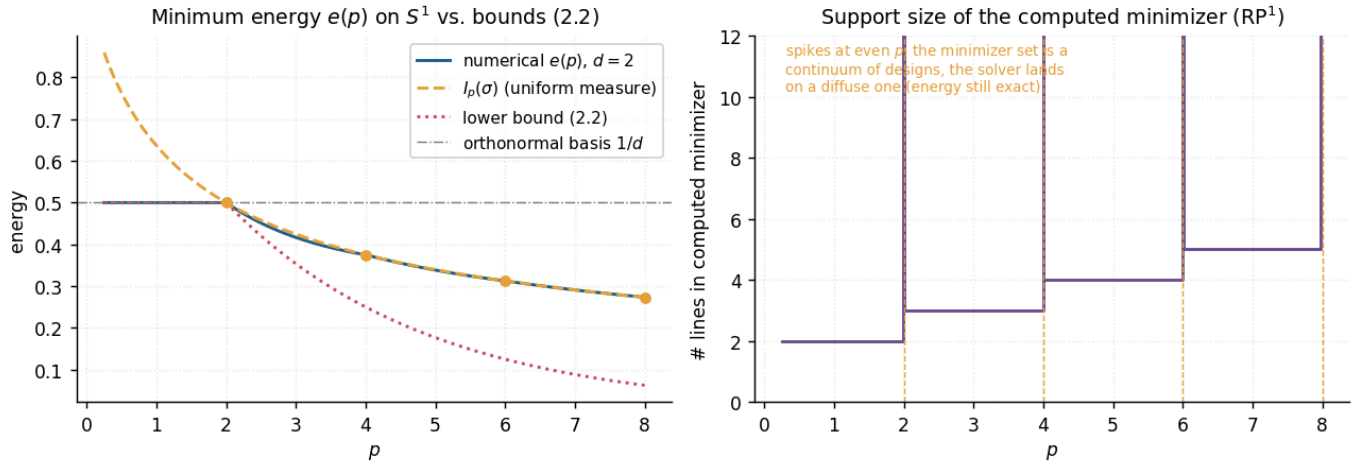
How the minimizers were computed

Measures are weighted atoms on RP^1 (angles in $[0, \pi)$) or lines in S^2. On S^1 a projected-gradient stage on a 180-point grid (16 random and N-gon starts, 1 200–2 500 iterations) is followed by L-BFGS refinement of positions and weights with analytic gradients, pruning and merging of atoms, repeated three times; on S^2 the same with a 700-point Fibonacci grid and ε-continuation for the non-smooth kernels at $p < 2$. Energies are compared with the closed forms $I_p(\sigma) = \Gamma((p+1)/2)/(\sqrt{\pi}\,\Gamma(p/2+1))$ on S^1 and $1/(p+1)$ on S^2. Exact or high-precision steps use sympy and mpmath; ranks in Lemma 3.1 are computed over $\mathbb{Q}$.

§1–2 Variational structure

✓ Energy bounds (2.2) and the even-exponent minimizer

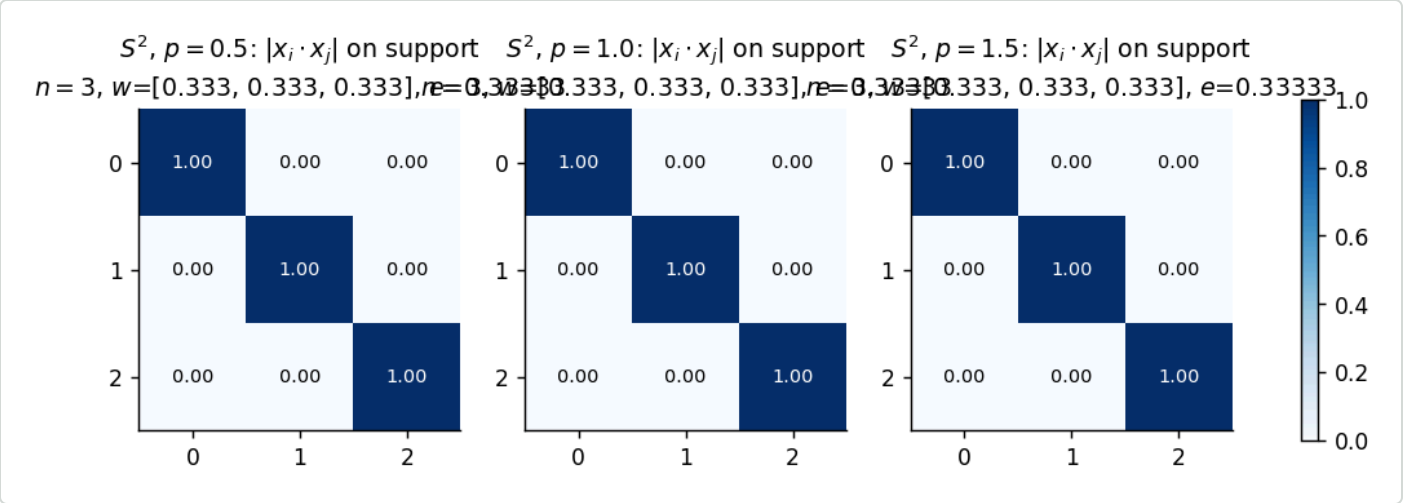
$e(p) \geq 1/d$ for $p \leq 2$ and $e(p) \geq d^{-p/2}$ for $p \geq 2$; for $p = 2k$ the rotation-invariant σ minimizes.



- Numerical e(p) on S^1 at 171 exponents in [0.25, 8]; minimum slack above the bound (2.2): -1.67e-16 (never negative).
- For $p \le 2$ the computed minimum is 1/2 to within 4.42e-05 (grid-limited at the non-smooth exponents).
- At p = 2, 4, 6, 8 the computed minimum equals $I_p(\sigma) = \Gamma((p+1)/2)/(\sqrt{\pi}\,\Gamma(p/2+1))$ to 0.00e+00, 5.55e-17, 1.67e-16.
- $I_2(\mu) = \mathrm{tr}(M_\mu^2)$ verified to 3.33e-16 on random measures; $\mathrm{tr}(M_\mu^2) - 1/d \ge 1.56e\text{-}02 > 0$.

✓ Proposition 2.2: 0 < p < 2 forces an orthonormal basis

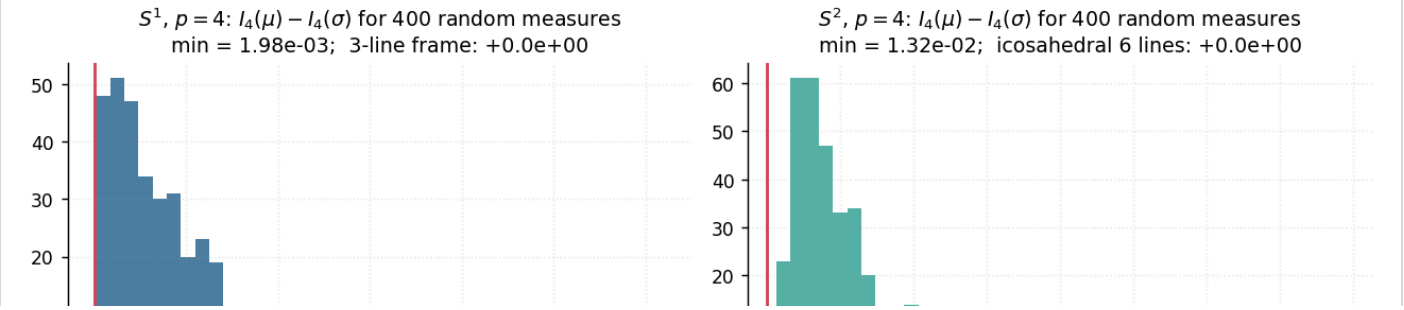
Every minimizer is carried by the d lines of an orthonormal basis, each with mass 1/d.

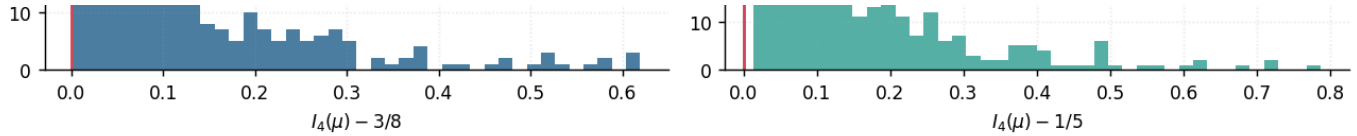


- S^2: for p = 0.5, 1, 1.5 the solver (grid stage + ε-continuation for the non-smooth kernel) returns exactly 3 lines; largest off-diagonal $|x_i \cdot x_j| = 2.24e\text{-}11$, $\|M_\mu - I/3\|_{\max} = 3.83e\text{-}08$.
- S^1: two lines at exactly 90° with masses 1/2 for all three exponents.

✓ Even exponents: averaging over O(d) is a projection

$I_{2k}(\mu) \ge I_{2k}(\sigma)$ for every μ; tight designs attain equality.

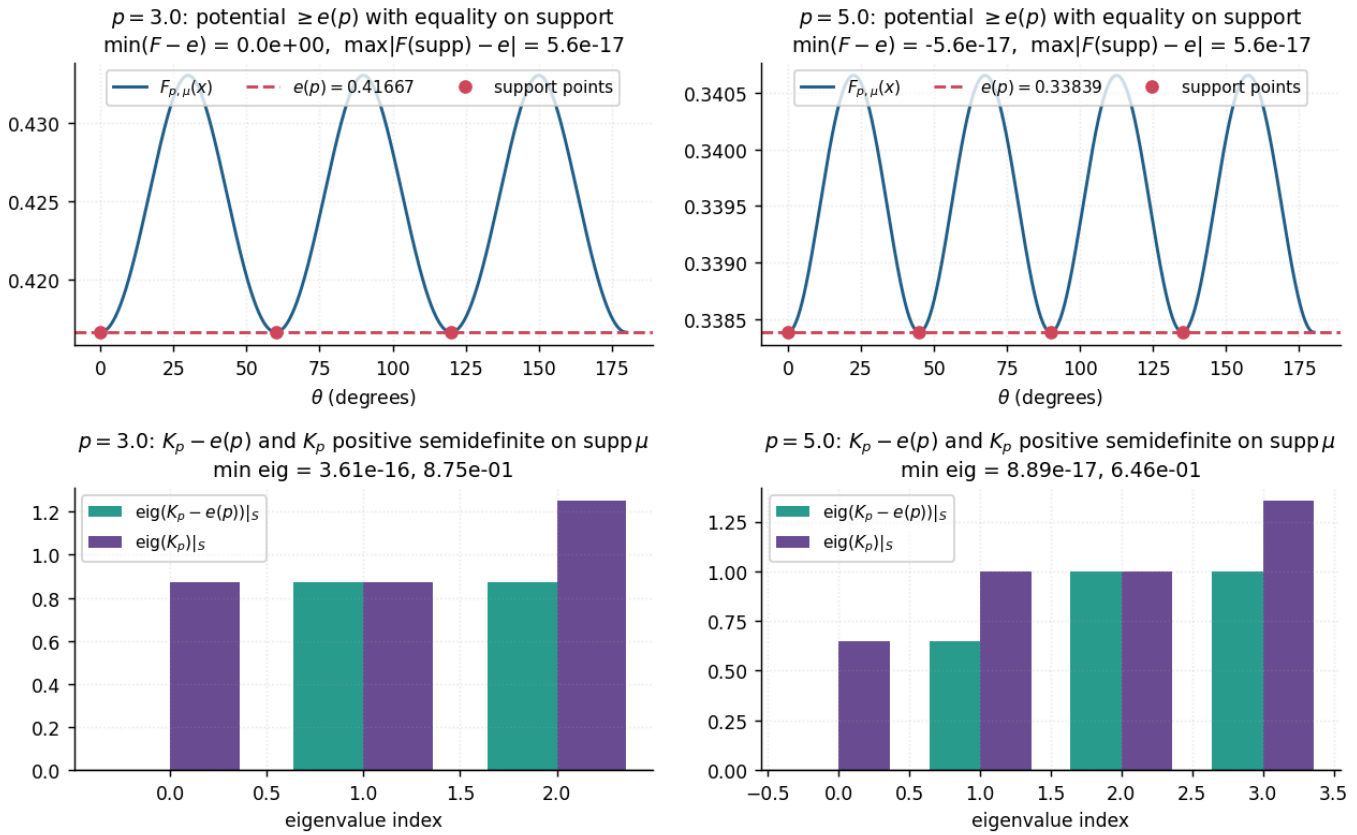




- 400 random measures on S^1 and S^2: $I_4(\mu) - I_4(\sigma) \geq$ 1.98e-03 and $\geq$ 1.32e-02.
- 3-line frame on S^1 and icosahedral 6 lines on S^2 reach 3/8 and 1/5 exactly (deviation 0.00e+00, 0.00e+00); the S^2 solver finds e(4) = 0.200000000000.

✓ Lemma 2.1: contact and centered positivity

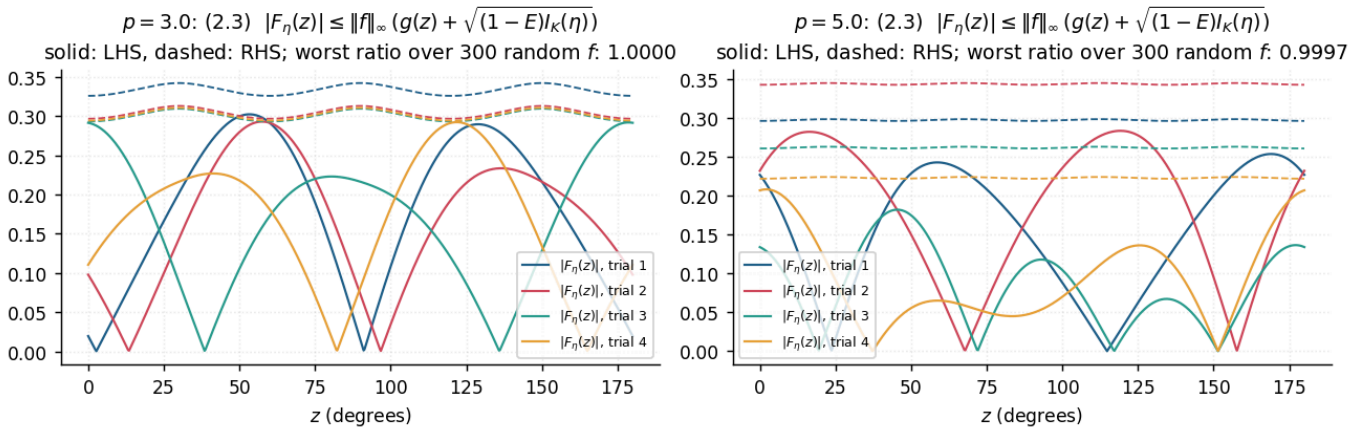
$F_\mu \geq E$ on X with equality on supp μ; $K - E$ (hence K, since $E \geq 0$) is positive semidefinite on supp μ.



- p = 3 (3 lines at 60°): $\min_{\text{circle}}$(F − e) = 0.00e+00, max |F − e| on support 5.55e-17, eig(K − e) ≥ 3.61e-16.
- p = 5 (4 lines at 45°): same checks give -5.55e-17, 5.55e-17, 8.89e-17.
- S^2, p = 3: the solver returns the 6 icosahedral lines (energy 0.2412022659, below $I_3(\sigma)$ = 1/4); F − e ≥ 8.65e-08 on 20 000 random points, eig(K − e) ≥ -4.44e-16.
- The minimizing support spans $\mathbb{R}^d$ in every computed case (rank of the support equals d).

✓ Proposition 2.3: the global variational bound (2.3)

$|F_\eta(z)| \le \|f\|_\infty (g(z) + \sqrt{((1-E)\, I_K(\eta))})$ for $\eta = f\mu$ with $\int f\, d\mu = 0$.

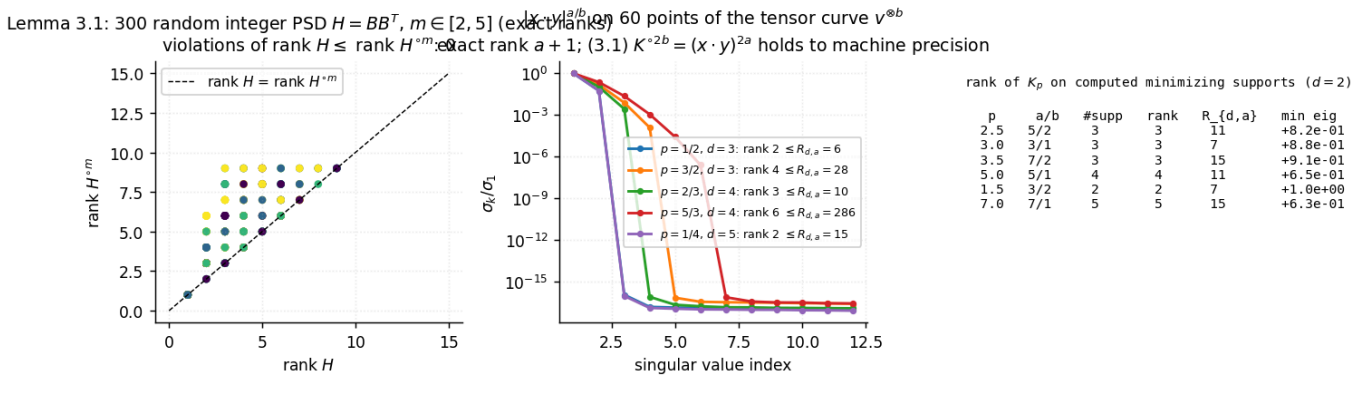


- 300 random zero-mean densities at p = 3 and p = 5: largest ratio $|F_\eta|$/bound = 0.999992 and 0.999672 — never above 1, and essentially tight.
- $I_K(\eta) \ge 0$ held in every trial (consistent with centered positivity).

§3 Rational exponents

✓ Lemma 3.1 and the rank bound (1.2)/(3.1)

rank $H \le$ rank $H^{\circ m}$ for PSD H; $K_p^{\circ 2b} = (x\cdot y)^{2a}$, so a PSD restriction of $K_{a/b}$ has rank $\le R_{d,a}$.



rank of K_p on computed minimizing supports ($d = 2$)

p	a/b	#supp	rank	R_{d,a}	min eig
2.5	5/2	3	3	11	+8.2e-01
3.0	3/1	3	3	7	+8.8e-01
3.5	7/2	3	3	15	+9.1e-01
5.0	5/1	4	4	11	+6.5e-01
1.5	3/2	2	2	7	+1.0e+00
7.0	7/1	5	5	15	+6.3e-01

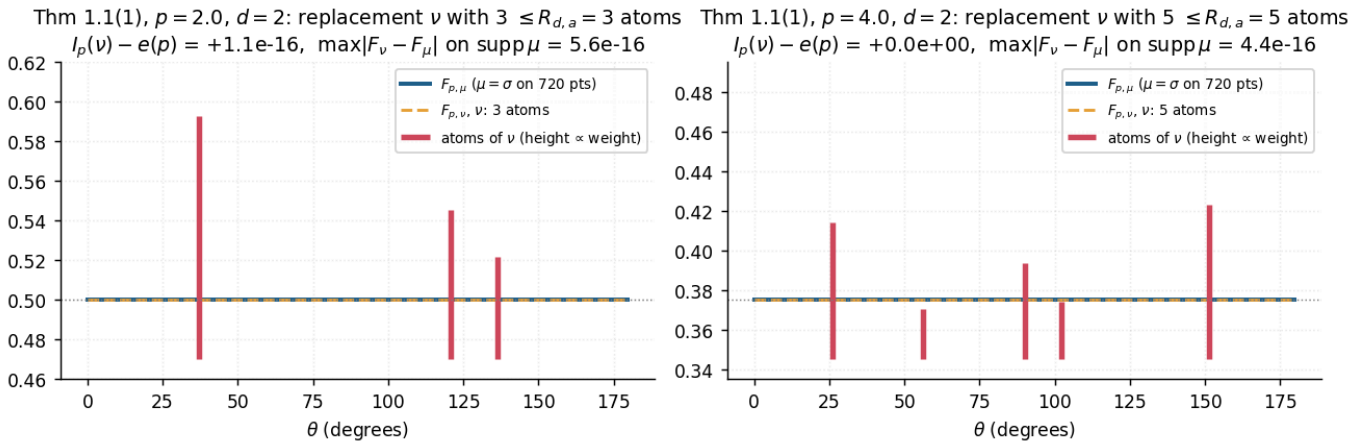
- 300 random integer Gram matrices with m ∈ [2,5], ranks computed exactly over $\mathbb{Q}$: 0 violations.
- On 60 points of the tensor curve $v^{\otimes b}$ the kernel $|x\cdot y|^{a/b}$ has exact numerical rank a+1 for

(a,b) = (1,2),(3,2),(2,3),(5,3),(1,4) — all ≤ $R_{d,a}$; identity (3.1) holds to 8.88e-16.

- On every computed minimizing support (p = 1.5, 2.5, 3, 3.5, 5, 7) the restricted kernel is positive definite with rank ≤ $R_{2,a}$.

✓ Theorem 1.1(1): finite minimizing replacement

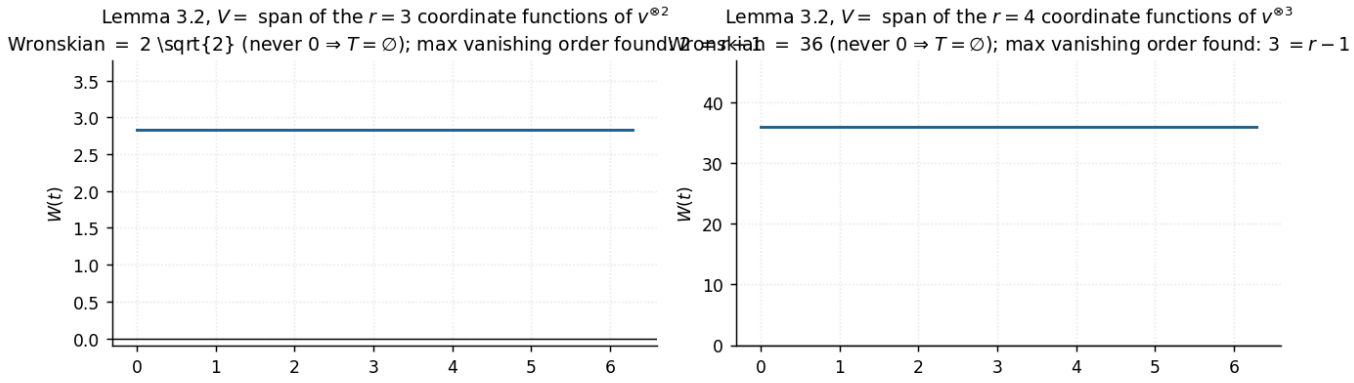
Affine Carathéodory in the finite-rank feature space gives ν with ≤ $R_{d,a}$ atoms, the same energy and the same potential on supp μ.



- μ = σ (720 points) at p = 2: a basic LP solution has 3 ≤ D_2 = 3 atoms, $I_2(\nu)$ − e = 1.11e-16, max $|F_\nu - F_\mu|$ on supp μ = 5.55e-16.
- p = 4: 5 ≤ D_4 = 5 atoms, $I_4(\nu)$ − e = 0.00e+00, potential match 4.44e-16; ⟨Φ(x), M⟩ = e on all of supp μ to 5.00e-16 (the affine-hyperplane step), $\|M\|^2$ = e to 4.44e-16.

✓ Lemma 3.2: finite-dimensional vanishing orders

The Wronskian of a basis of V is not identically zero, so off a finite set T every nonzero f ∈ V vanishes to order ≤ r−1.

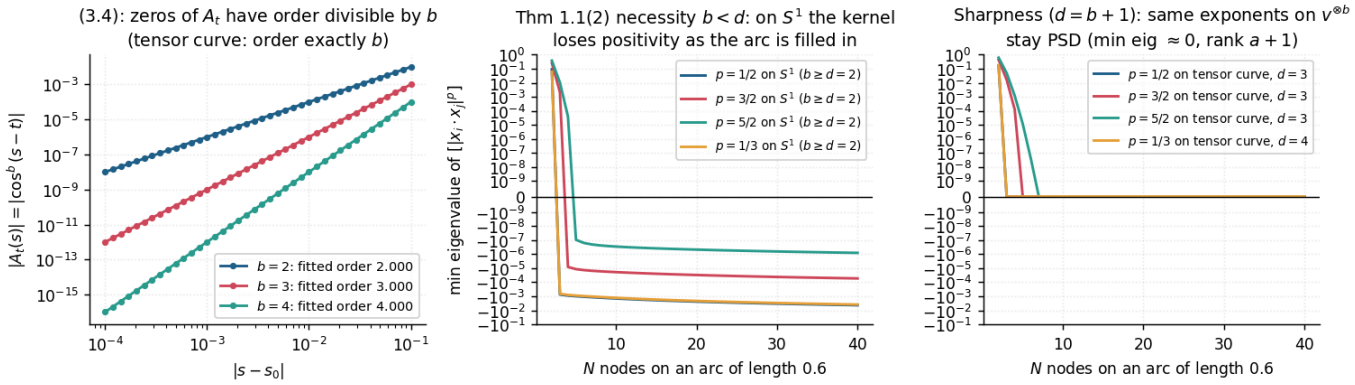


t t

- For the coordinate functions of $v^{\otimes 2}$ (r = 3) and $v^{\otimes 3}$ (r = 4), sympy gives the exact Wronskians 2*sqrt(2) and 36: nonvanishing constants, so T = ∅.
- Vanishing orders at 0 over 200 random elements: ≤ 2 (b=2) and ≤ 3 (b=3); the bound r−1 is attained by $\sin^b t$.

✓ (3.3)–(3.4) and sharpness of the denominator obstruction

$B_t^{2b} = A_t^{2a}$ forces $b \mid \mathrm{ord}_z A_t$; $b \geq d$ is impossible for an infinite PSD support, and $b = d-1$ occurs (tensor curve).

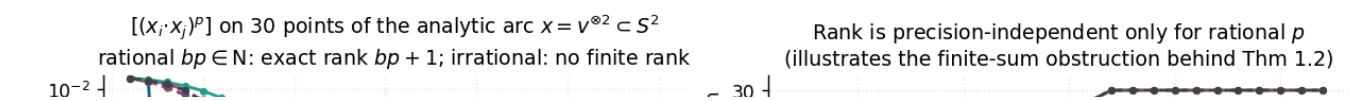


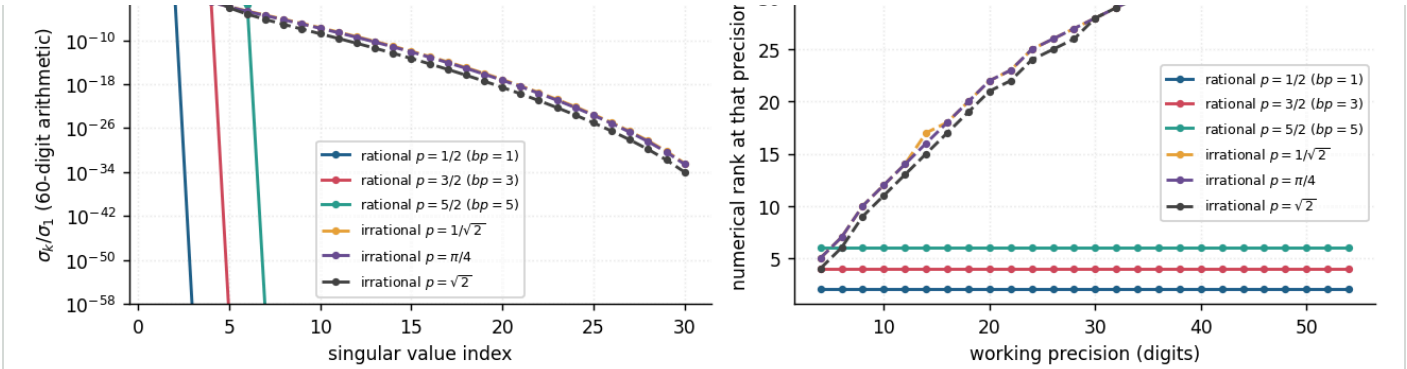
- Symbolic identity $B_t^{2b} - A_t^{2a} = 0$ for (a,b) = (1,2),(3,2),(2,3); fitted zero orders of A_t: 2.000, 3.000, 4.000 = b.
- S^1 (b ≥ d): for p = 1/2, 3/2, 5/2, 1/3 the kernel matrix on 40 nodes of an arc has min eigenvalue -4.33e-03, -5.33e-05, -8.06e-07, -3.89e-03 — positivity is lost as the arc fills.
- Same exponents on $v^{\otimes b} \subset S^b$: min eigenvalue ≈ 0 (-8.25e-15 etc.) and rank a+1, matching the paper's sharpness example.

§4 Analytic rigidity and great circles

◐ Theorem 1.2: finite rank ⇔ finite support at irrational p

The Cheng–Di Scala–Yuan finite-sum theorem is the analytic input; it cannot be verified numerically, but its mechanism can be shown.

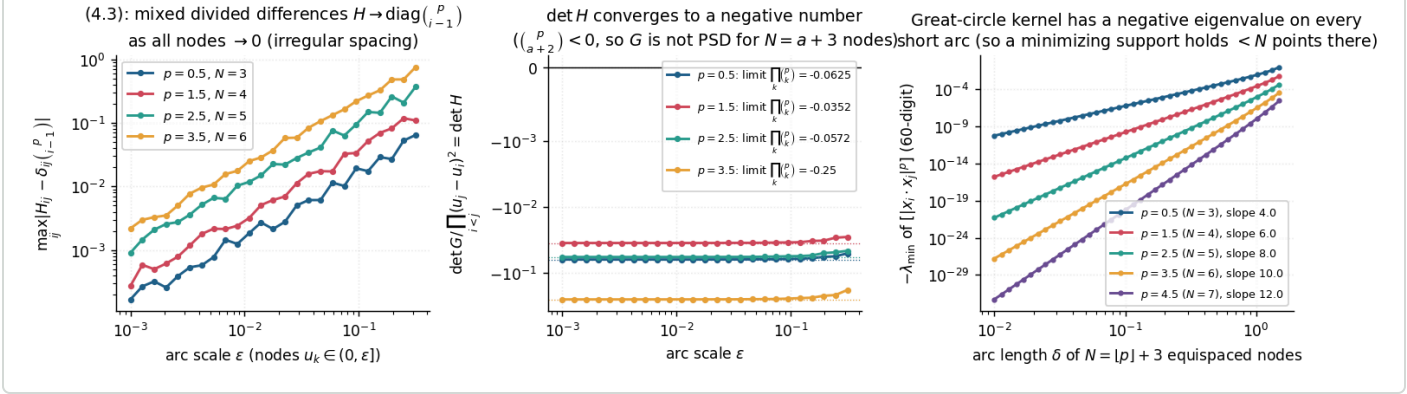




- In 60-digit arithmetic on 30 points of the analytic arc $v^{\otimes 2}$: for p = 1/2, 3/2, 5/2 the rank is exactly bp+1 = 2, 4, 6 at every working precision; for p = 1/√2, π/4, √2 the numerical rank grows by one with every two added digits, with no exact drop.
- Curve selection, Noetherianity and the Umehara-algebra argument are structural and left to the proof text; this figure only shows that a rational exponent and nothing else produces a finite Gram representation along an analytic arc.

✓ Theorem 4.1 (noninteger p): divided differences and the sign obstruction

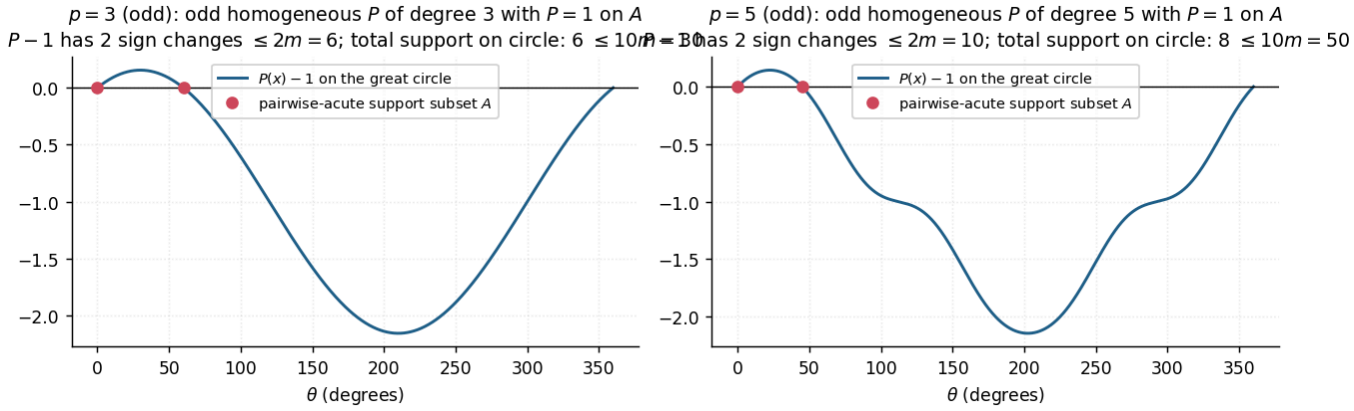
$\det G = \prod(u_j - u_i)^2 \det H$ with $H \to \mathrm{diag}(\mathrm{binom}(p, i-1))$; $\mathrm{binom}(p, a+2) < 0$, so $N = \lfloor p \rfloor + 3$ nodes in a short arc contradict PSD.



- Mixed divided differences of $(1+uv)^p$ computed in 60-digit arithmetic with irregular nodes: max |H − diag| = 9.08e-04 at $\varepsilon = 10^{-3}$ (p = 2.5), and det H → $\prod_k \mathrm{binom}(p,k)$ = -0.05722 (computed -0.05722).
- binom(p,0..a+1) > 0 and binom(p,a+2) < 0 confirmed for p = 0.5, 1.5, 2.5, 3.5.
- Diagonal congruence between $(1+u_iu_j)^p$ and the sphere kernel verified to 5.55e-16; inertia (one negative eigenvalue) agrees.
- The true great-circle kernel has a negative eigenvalue on every arc tested; $-\lambda_{\min} \propto \delta^{2(N-1)}$ (fitted slopes 4, 6, 8, 10, 12), exactly the Vandermonde factor in (4.3).

✓ Theorem 4.1 (odd p): the trigonometric zero count

On a pairwise-acute subset the Riesz representer is an odd homogeneous P of degree m; P − 1 has ≤ 2m zeros on a great circle, giving 10m with five arcs.

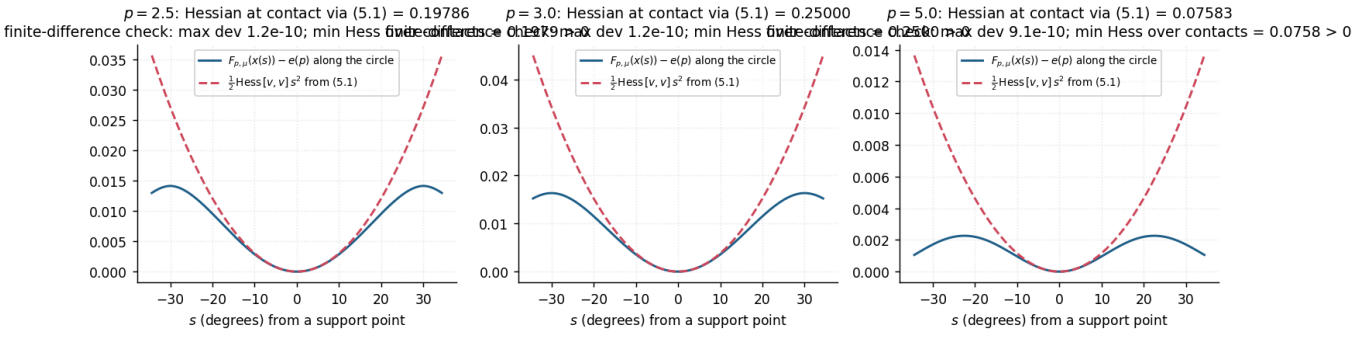


- p = 3: P = 1 on A to 1.11e-16; P − 1 changes sign 2 ≤ 6 times; the support meets the circle in 6 ≤ 30 points.
- p = 5: 2 ≤ 10 sign changes, 8 ≤ 50 support points; the functional ℓ is bounded (eig(K − e) on A ≥ 0.500).

§5 Contact curvature

✓ The Hessian formula (5.1) and strict positivity at contacts

Hess $F_{p,\mu}(x)[v,v] = p(p-1)\int |x\cdot y|^{p-2}(v\cdot y)^2\, d\mu - p\, e(p) \geq 0$ at contacts.

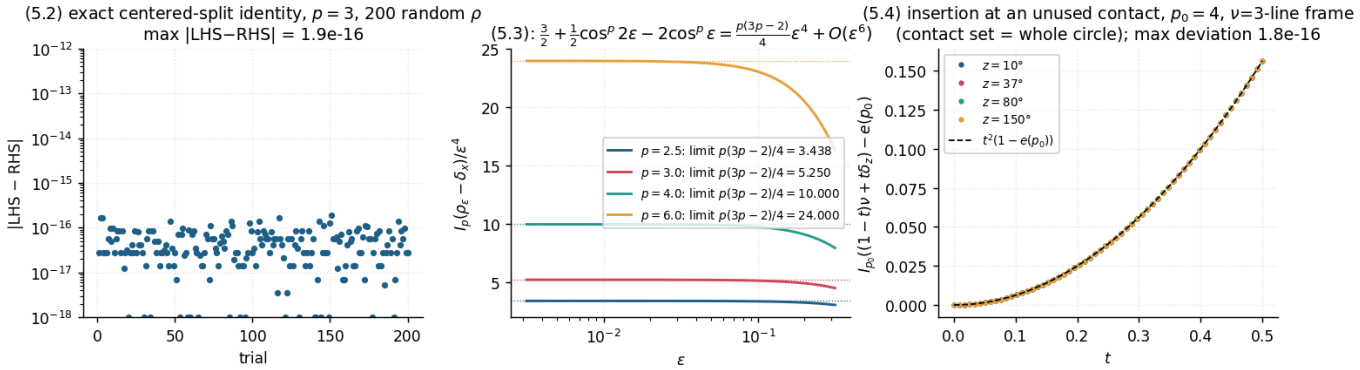


- S^1, p = 2.5 / 3 / 5: formula vs fourth-order finite differences agree to 1.22e-10, 1.23e-10, 9.13e-10; minimum Hessian over contacts 0.1979, 0.2500, 0.0758 (strictly positive: the hypothesis of Thm 1.3 holds at these exponents).

- S^2, p = 3, icosahedral minimizer, two tangent directions and their bisector: formula 0.170820 vs finite difference 0.170820.

✓ The exact centered split (5.2)–(5.4)

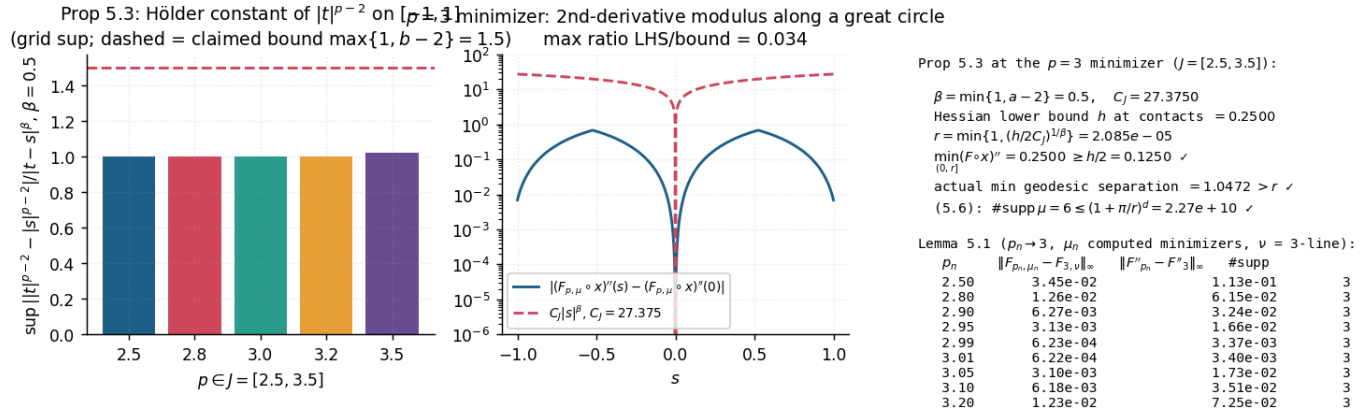
Identity (5.2) is exact; $I_p(\rho_\varepsilon - \delta_x) = p(3p-2)\varepsilon^4/4 + O(\varepsilon^6)$; insertion at an unused contact costs $t^2(1 - e(p_0))$.



- (5.2): 200 random replacements ρ at p = 3, max |LHS − RHS| = 1.94e-16.
- (5.3): $I_p(\rho_\varepsilon - \delta_x)/\varepsilon^4$ at ε = 0.01 equals 5.24924 vs p(3p−2)/4 = 5.25 (p = 3), 23.9906 vs 24.0 (p = 6); the ε^6 coefficient stays bounded.
- (5.4): at p_0 = 4 the 3-line frame has constant potential (range 3.33e-16), so every point is an unused contact; insertion cost matches t^2(1−e) to 1.80e-16 at four test points.

✓ Lemma 5.1 and Proposition 5.3: continuity, Hölder constant, separation, packing

Potentials converge in C^2 along $p_n \to p_0$; $|t|^{p-2}$ is β-Hölder with constant ≤ max{1, b−2}; contacts are separated by r and (5.6) bounds the support.



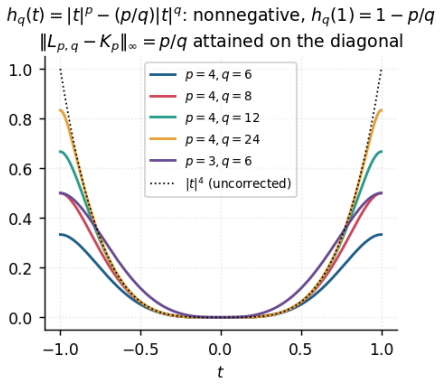
3.50 3.00e-02 1.96e-01 3

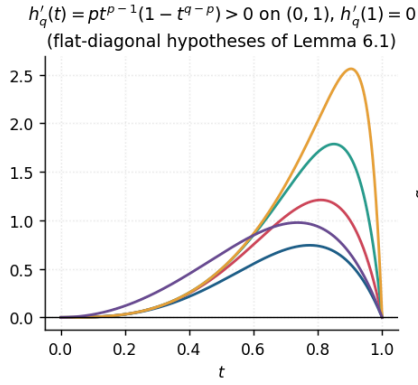
- Hölder constant of $|t|^{p-2}$ on $[-1,1]$ with $\beta = 0.5$ for $p \in [2.5, 3.5]$: at most $1.0219 \le 1.5$.
- Along a great circle through a support point of the p = 3 minimizer, $|(F\circ x)''(s) - (F\circ x)''(0)| \le C_J|s|^{\beta}$ with ratio at most 0.034.
- With h = 0.2500 the bound gives r = 2.09e-05; $(F\circ x)'' \ge h/2$ on (0, r] (min 0.2500); actual separation 1.0472; (5.6) holds with large slack, as expected from a conditional estimate.
- Lemma 5.1: $\|F_{p_n,\mu_n} - F_{3,\nu}\|_\infty$ and the second-derivative error both decrease linearly in $|p_n - 3|$ (table in the figure); the support count stays at 3, consistent with Proposition 5.2.

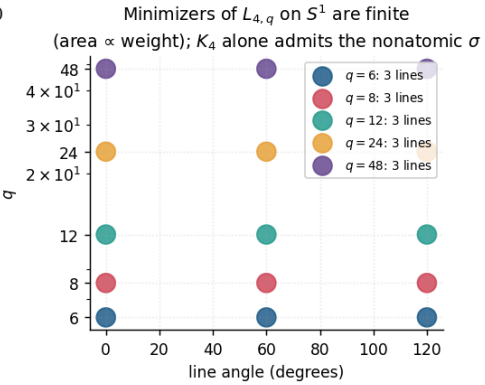
§6 Finite-support regularization

✓ Theorem 1.4 and Lemma 6.1: the flat-diagonal correction

$h_q(t) = |t|^p - (p/q)|t|^q$ is nonnegative with $h_q(1) = 1-p/q$, $h_q'(1) = 0$, $h_q' > 0$ on (0,1); $\|L_{p,q} - K_p\|_\infty = p/q$; every $L_{p,q}$-minimizer is finite.



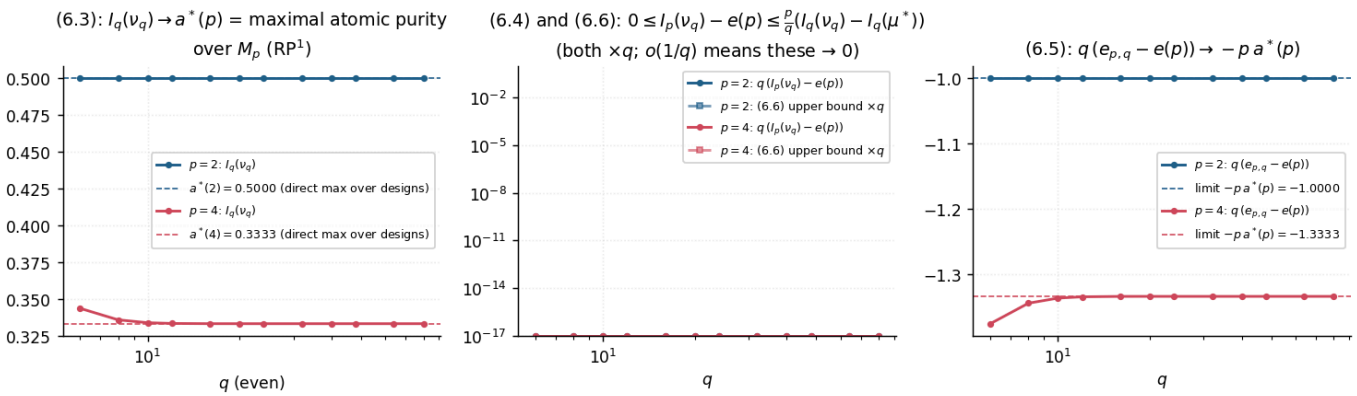




- All hypotheses checked on a 2001-point grid for (p,q) = (4,6),(4,8),(4,12),(4,24),(3,6); e.g. (4,6): min h = 0.0, h(1) = 0.333333, sup|L−K| = 0.666667 = p/q attained at |t| = 1, h''(1) = -8.0 = p(p−q).
- Minimizers of $L_{4,q}$ on S^1 for q = 6, 8, 12, 24, 48: in every case the 3-line frame (finite), with the corrected kernel positive definite on the support (min eig 0.2812); the uncorrected K_4 admits the nonatomic σ.

✓ Proposition 6.2: maximal-purity selection

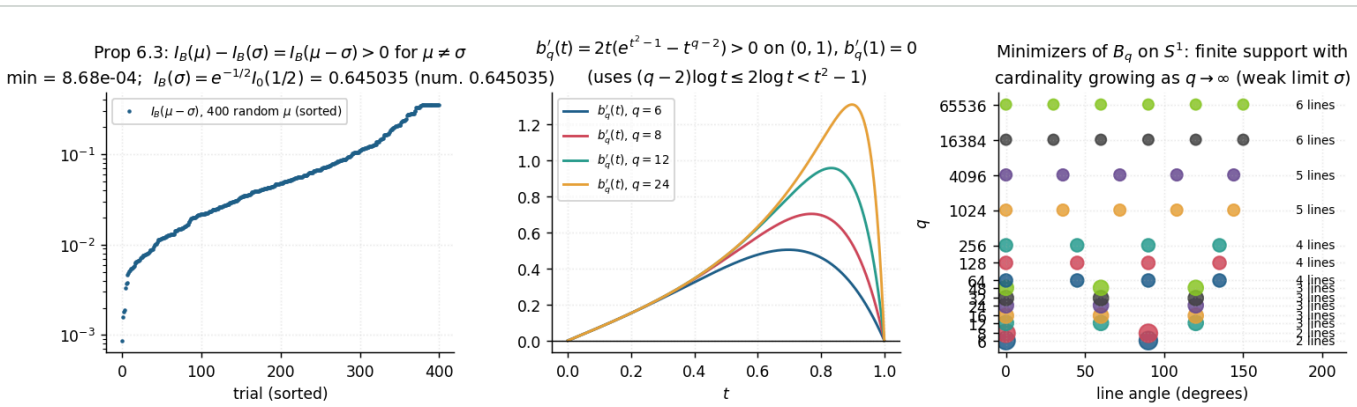
$I_q(\nu_q) \to a^*(p)$, $I_p(\nu_q) - e(p) = o(1/q)$, $e_{p,q} = e(p) - (p/q)a^*(p) + o(1/q)$.



- $a^*(2) = 0.500000$ and $a^*(4) = 0.333333$ from direct purity maximization over designs on RP^1 (orthonormal basis and 3-line frame).
- For $q = 6 \ldots 80$, $I_q(v_q)$ converges to exactly these values, $I_p(v_q) - e(p)$ is 0 to machine precision, and $q(e_{p,q} - e(p)) \to -p\,a^*(p)$ (−1 and −4/3); the sandwich (6.6) holds at every q.

✓ Proposition 6.3: the rotationally invariant counterexample

$B(x,y) = \exp((x\cdot y)^2 - 1)$ has the unique, nonatomic minimizer σ; every B_q-minimizer is finite, converges weakly to σ, with support size → ∞.

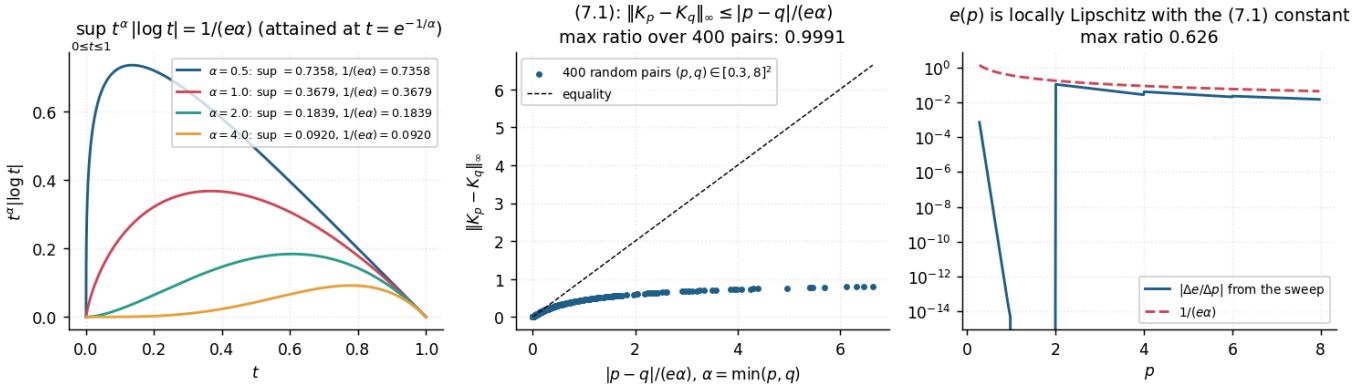


- $I_B(\sigma) = e^{-1/2} I_0(1/2) = 0.64503527$ matches the numerical value; $I_B(\mu) - I_B(\sigma) = I_B(\mu - \sigma)$ to 9.44e-16 and is ≥ 8.68e-04 > 0 over 400 random μ.
- $b_q \ge e^{-1} - 2/q$, $b_q(1) = 1 - 2/q$, $b_q'(1) = 0$, $b_q' > 0$ on (0,1) and $(q-2)\log t - (t^2-1) < 0$ verified for q = 6, 8, 12, 24.
- B_q-minimizers on S^1 are regular N-line frames with N = 2, 2, 3, 3, 3, 3, 3, 4, 4, 4, 5, 5, 6, 6 for q = 6, 8, 12, 16, 24, 32, 48, 64, 128, 256, 1024, 4096, 16384, 65536; energies rise toward $I_B(\sigma)$ (0.6450306 at q = 65536). The growth is real but only logarithmic in q — worth a remark in the text, since the proposition promises divergence without a rate.

§7 Approximation, selection, recovery

✓ The exponent-Lipschitz estimate (7.1)

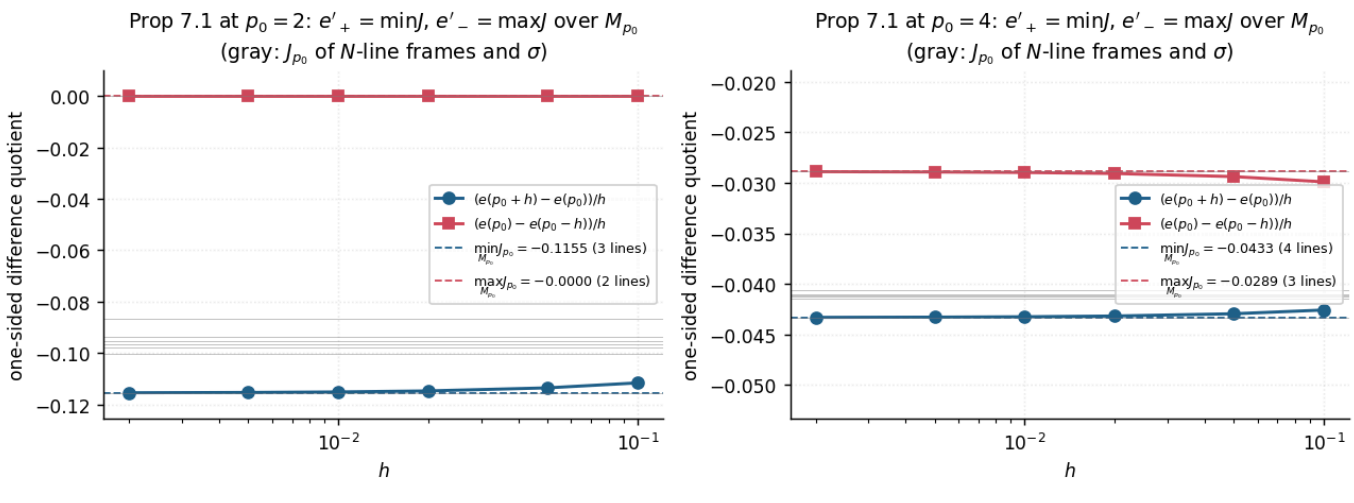
$\|K_p - K_q\|_\infty \le |p-q|/(e\alpha)$ because $\sup_{0\le t\le 1} t^\alpha|\log t| = 1/(e\alpha)$; $e(p)$ is locally Lipschitz.



- sup $t^\alpha|\log t| = 0.36787944$ vs $1/(e\alpha) = 0.36787944$ at $\alpha = 1$, attained at $t = e^{-1/\alpha}$; same for $\alpha = 0.5, 2, 4$.
- 400 random exponent pairs in $[0.3, 8]$: max ratio $\|K_p - K_q\|_\infty$/bound = 0.9991; difference quotients of the computed $e(p)$ stay below $1/(e\alpha)$ (max ratio 0.626).

✓ Proposition 7.1: one-sided selection and the derivative formula

Limits from above minimize J_p over M_p, limits from below maximize it, and $e'_\pm(p)$ are these extreme values.



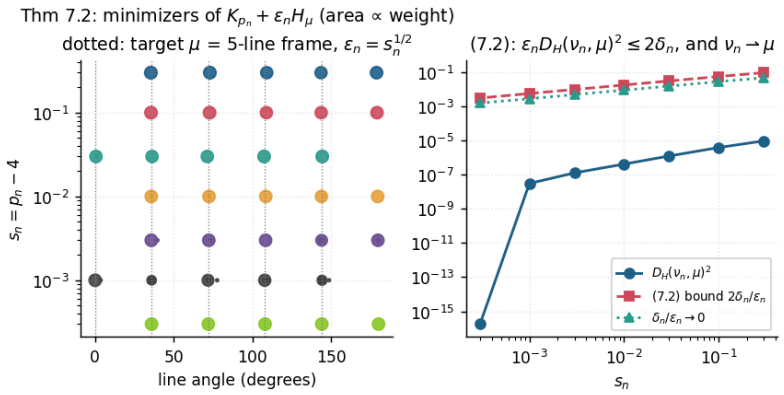
- $p_0 = 2$: right difference quotients → -0.11544 at h = 0.002 vs min J_2 = -0.11552 (3-line

frame, the minimizer found at every p slightly above 2); left quotients are exactly 0 = max J_2 (orthonormal basis).

- $p_0 = 4$: right → -0.04331 vs min J_4 = -0.04332 (4-line frame); left → -0.02890 vs max J_4 = -0.02888 (3-line frame). The extremes were found by constrained optimization over 4-designs and coincide with the frames selected by the exponent sweep.
- σ and the 5-, 6-, 7-, 8-line frames all minimize I_4 but have intermediate J_4 (-0.0412, -0.0406, …): none of them is a one-sided limit, exactly as the proposition predicts.

✓ Theorem 7.2 and Corollary 7.3: target-centered recovery

With H_μ the centered dense-family kernel, every minimizer of $K_n + \varepsilon_n H_\mu$ converges weakly to the prescribed μ, and $\varepsilon_n D_H(\nu_n,\mu)^2 \le 2\delta_n$.



Recovery data (μ = 5-line frame, $p_n = 4 + s_n$, $\varepsilon_n = \sqrt{s_n}$):

s_n	δ_n	D_H^2	$\varepsilon_n D_H^2/2\delta_n$	mass≤2° of μ	#atoms ($w > 10^{-3}$)
0.3000	2.7e-02	9.6e-06	9.9e-05	1.00000	5
0.1000	9.1e-03	3.9e-06	6.9e-05	1.00000	5
0.0300	2.7e-03	1.3e-06	4.0e-05	1.00000	5
0.0100	9.2e-04	4.2e-07	2.3e-05	1.00000	8
0.0030	2.8e-04	1.3e-07	1.3e-05	0.99921	8
0.0010	9.2e-05	3.1e-08	5.4e-06	0.99921	11
0.0003	2.8e-05	1.8e-16	5.6e-14	1.00000	5

$I_{H_\mu}(\rho) = D_H(\rho,\mu)^2$ check: -2.8e-17
H_μ PSD on random points: min eig 9.8e-04
Target is in M_4: $I_4(\mu) - e(4) = 0.0e+00$
Support cardinality is NOT controlled by Thm 7.2 (as the paper states): near-degenerate K_{p_n} lets atoms split/lose mass near the target lines; weak convergence still holds.

- Target μ = 5-line frame (a minimizer of K_4 that neither one-sided limit selects); $I_{H_\mu}(\rho) = D_H(\rho,\mu)^2$ verified to -2.78e-17, H_μ PSD.
- $p_n = 4 + s_n$, $\varepsilon_n = \sqrt{s_n}$, δ_n computed exactly: D_H^2 falls from 9.64e-06 to 1.78e-16, the ratio $\varepsilon_n D_H^2/2\delta_n$ never exceeds 9.92e-05, and the mass within 2° of the target lines is ≥ 0.99921.
- The number of atoms is not controlled (some runs split tiny mass off the target lines), which is precisely the separation the paper insists on: recovery gives weak convergence, Corollary 7.3 needs a support bound as an extra input.

Figures and numbers were produced from the uploaded manuscript (15 pp., v1 of 26 Sep 2026). "Verified numerically" means the stated identity or inequality held in every test performed, at the precision quoted; it is evidence for, not a replacement of, the proofs.